

UART Transmitter (UART_TX)

RTL Design & Functional Verification Report

1. Introduction

UART (Universal Asynchronous Receiver/Transmitter) is a full-duplex serial communication block. This report covers only the transmitter half (UART_TX), which converts an 8-bit parallel byte into a serial bit stream according to the frame format defined in the specification document.

The transmitter is configurable at run time through two static configuration inputs: PAR_EN (enable/disable a parity bit) and PAR_TYP (choose even or odd parity).

1.1 Block Interface

Port	Dir	Width	Description
CLK	In	1	UART TX clock (also the bit-rate clock - no baud generator is required)
RST	In	1	Asynchronous, ACTIVE LOW reset
PAR_TYP	In	1	0 = even parity, 1 = odd parity (configuration)
PAR_EN	In	1	0 = parity disabled, 1 = parity enabled (configuration)
P_DATA	In	8	Parallel data byte to transmit
DATA_VALID	In	1	1-clock pulse indicating P_DATA is valid
TX_OUT	Out	1	Serial output line (idles HIGH)
Busy	Out	1	HIGH while a frame is being transmitted

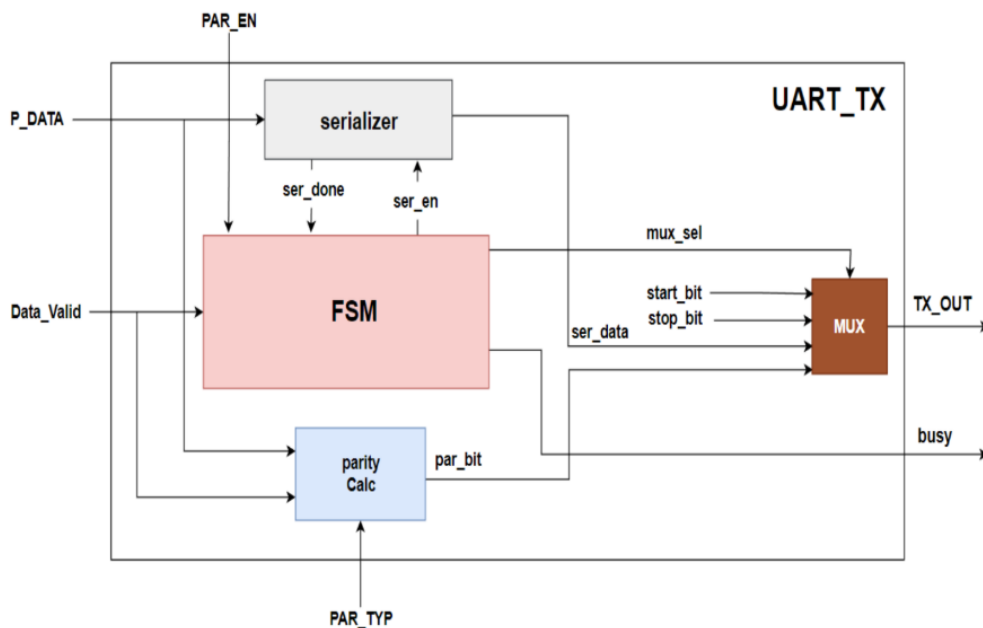
1.2 Frame Formats

Every frame starts with a start bit (0) and ends with a stop bit (1). Between them: 8 data bits, LSB first, followed optionally by 1 parity bit.

- Parity enabled + even (PAR_EN=1, PAR_TYP=0): START - D0..D7 - EVEN_PARITY - STOP
- Parity enabled + odd (PAR_EN=1, PAR_TYP=1): START - D0..D7 - ODD_PARITY - STOP
- Parity disabled (PAR_EN=0) : START - D0..D7 - STOP

2. Architecture / Block Diagram

The design is partitioned exactly as recommended in the spec, into four cooperating blocks controlled by one FSM:



Block	Responsibility
FSM (fsm.v)	Sequences the frame: IDLE -> START -> DATA(x8) -> [PARITY] -> STOP -> IDLE. Drives busy, load, ser_en and mux_sel.
Serializer (serializer.v)	Latches the byte on 'load' and exposes it bit-by-bit (LSB first) on ser_data, one new bit per clock while ser_en is high. Flags ser_done on the last bit.
Parity Calc (parity_calc.v)	Combinationally computes the parity bit from the latched byte, for both even and odd modes.
MUX (tx_mux.v)	Chooses what actually appears on TX_OUT (0, 1, ser_data or par_bit) based on the FSM's current state (mux_sel).

2.1 Why CLK = bit clock

The spec's block diagram has no baud-rate generator, so CLK IS the bit-rate clock: every state lasts exactly 1 clock cycle, except DATA, which lasts exactly 8 cycles (one per data bit).

3. RTL Implementation

Full source files are provided alongside this report (serializer.v, parity_calc.v, tx_mux.v, fsm.v, uart_tx.v). Key design decisions:

- Reset: RST is asynchronous and active LOW, per spec ('negedge RST' in every sequential always block).
- Data capture: P_DATA is only sampled while in IDLE and DATA_VALID is high; a data_snapshot register also freezes the byte for the parity calculator, and PAR_EN is latched into par_en_latched at that same moment so a mid-frame configuration change cannot corrupt an in-flight frame.
- Busy rejection: while Busy = 1, the FSM is not in IDLE, so any DATA_VALID pulse is ignored - matching the spec requirement.
- Serializer: instead of a physical shift register, a 3-bit index (bit_index) walks across the byte (LSB first); this is functionally identical to shifting but easier to read and debug.
- mux_sel encoding is shared between fsm.v and tx_mux.v (0=IDLE, 1=START, 2=DATA, 3=PARITY, 4=STOP) so the FSM's state register doubles directly as the mux select.

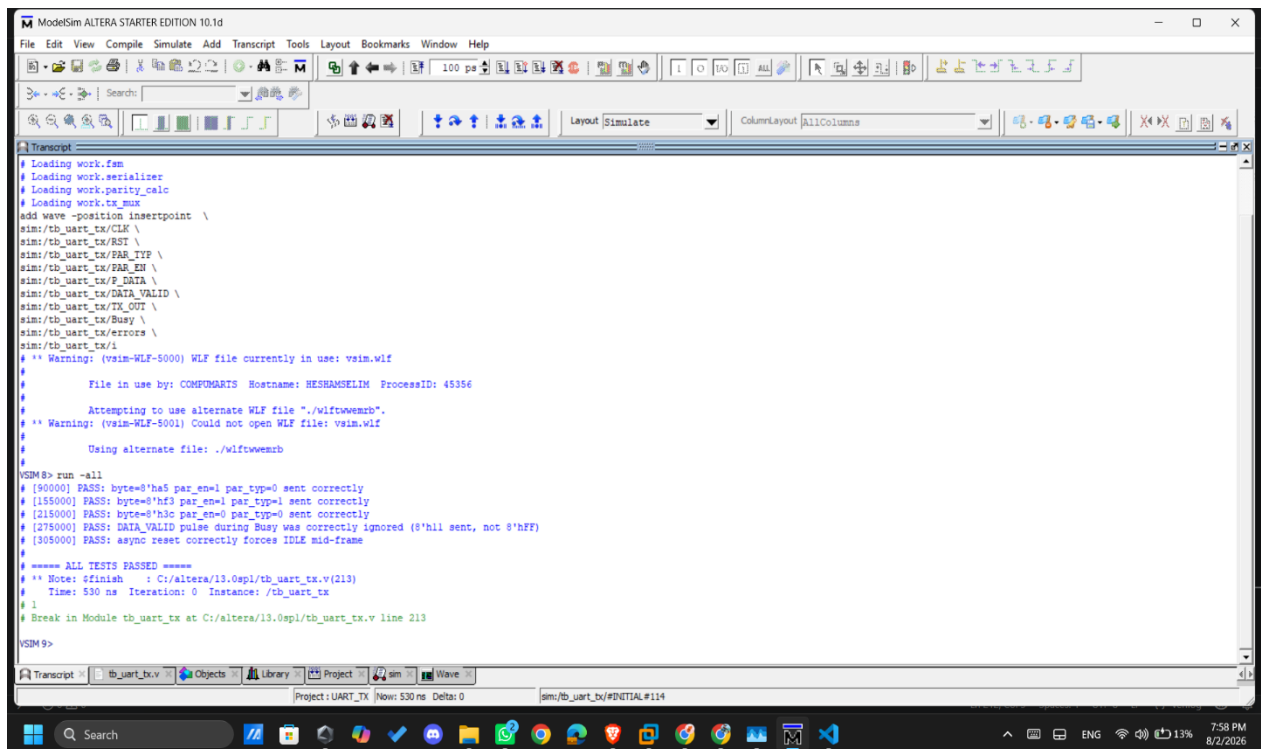
4. Testbench

The testbench (tb_uart_tx.v) drives a 200 MHz clock (5 ns period, generated with #2.5 CLK = ~CLK) and self-checks every bit of every transmitted frame against the expected value computed live in the testbench (no golden vectors needed).

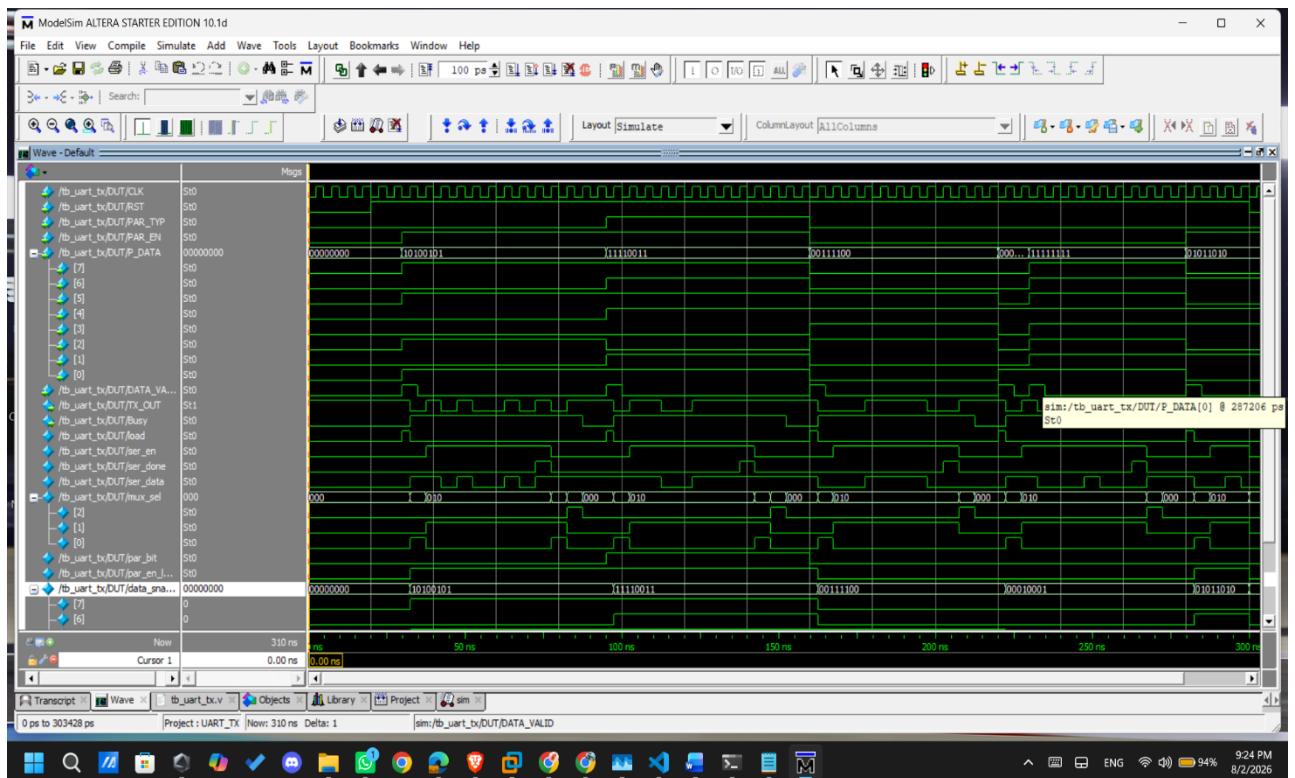
4.1 Test Cases

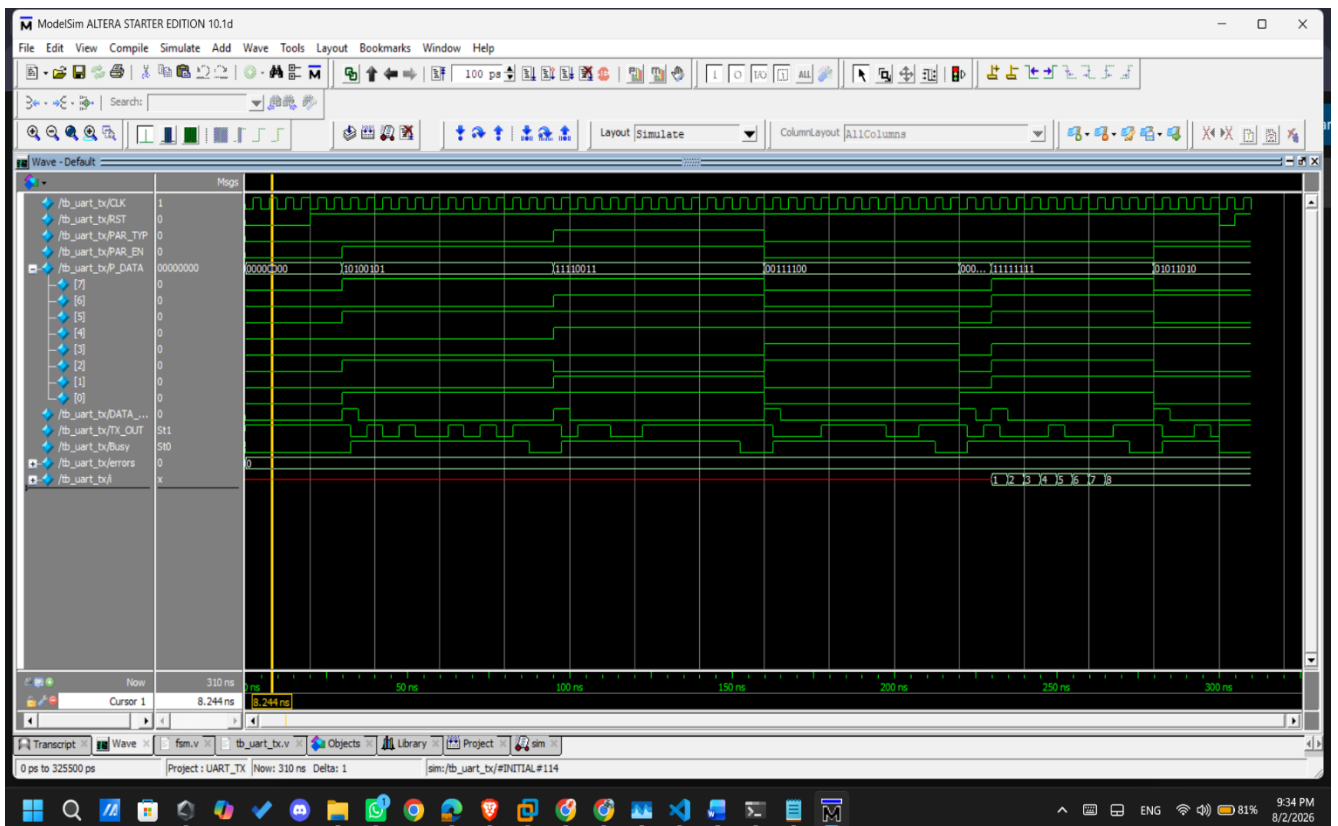
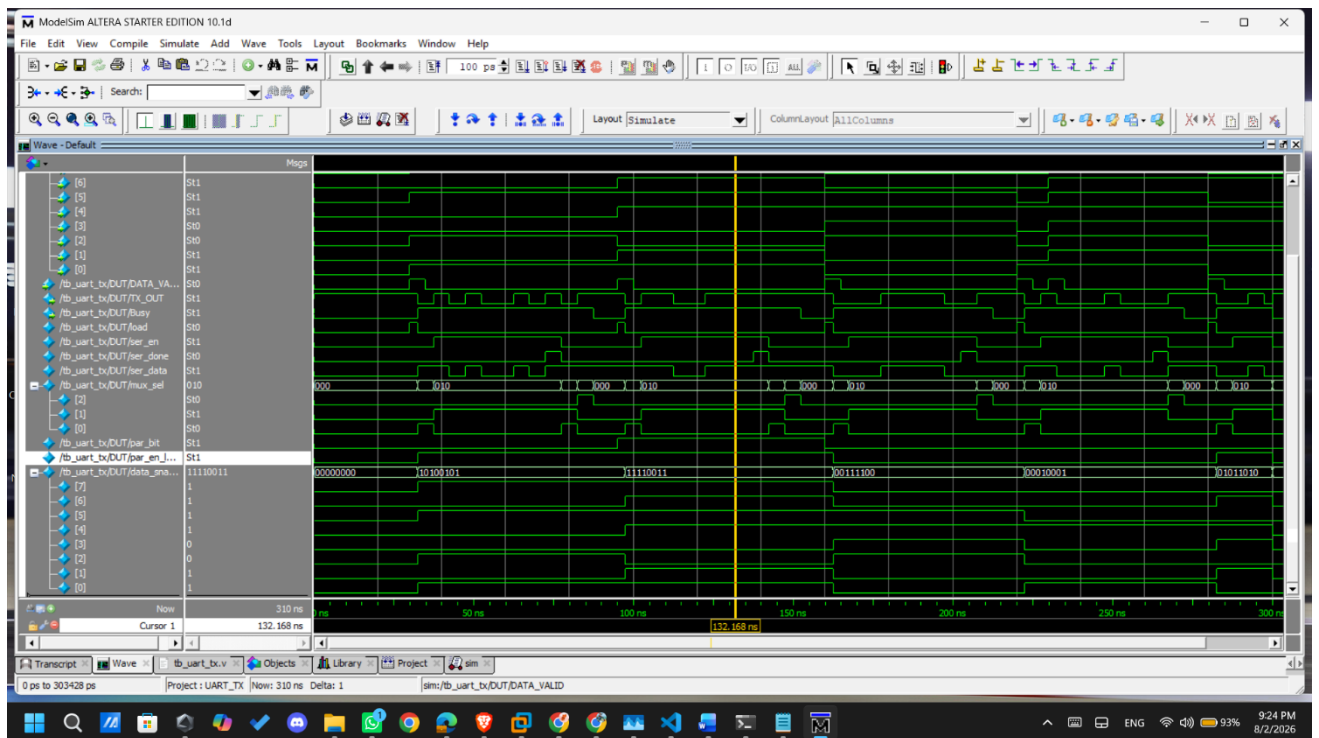
#	Scenario	Byte	PAR_EN	PAR_TYP	Checks
1	Parity enabled, even	8'hA5	1	0	Start, 8 data bits, even parity bit, stop, return to IDLE
2	Parity enabled, odd	8'hF3	1	1	Start, 8 data bits, odd parity bit, stop, return to IDLE
3	Parity disabled	8'h3C	0	-	Start, 8 data bits, stop (no parity bit), return to IDLE
4	DATA_VALID while Busy	8'h11 then 8'hFF ignored	0	-	Confirms the second byte (0xFF) is dropped and 0x11 is transmitted intact
5	Async reset mid-frame	8'h5A	1	0	RST asserted during the DATA state forces immediate return to IDLE (TX_OUT=1, Busy=0)

4.2 Simulation Log



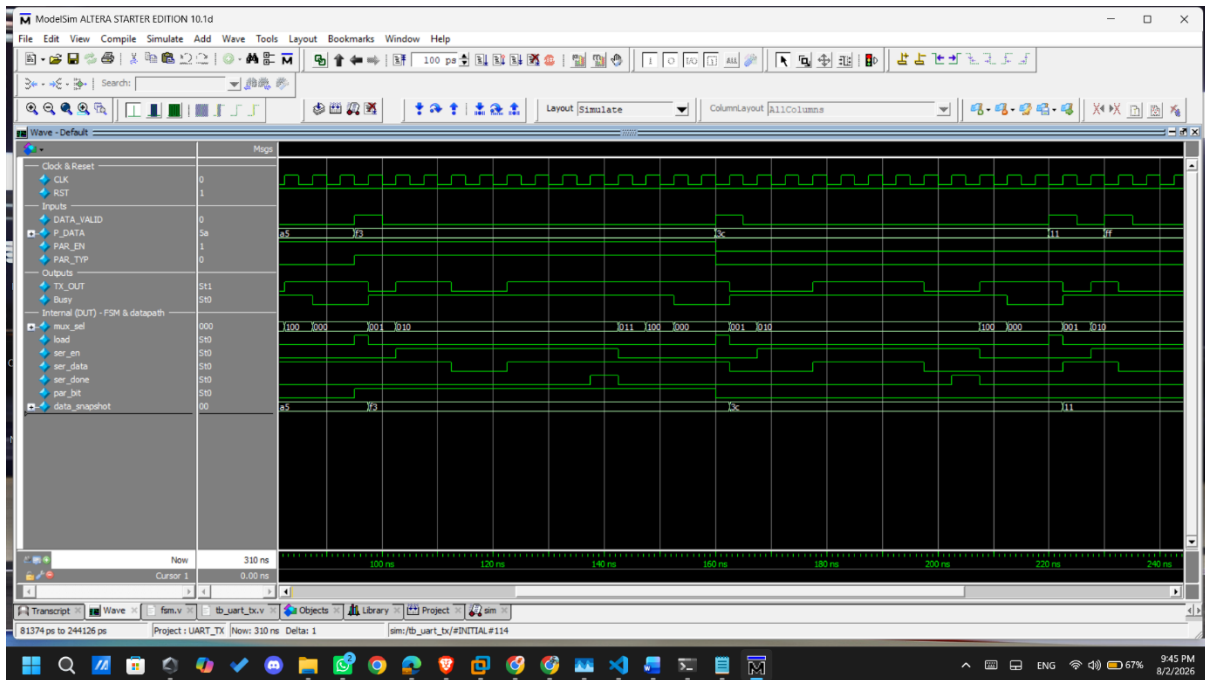
4.3 Waveform - Frame for P_DATA = 8'hA5 (parity enabled, even)





Reading the waveform: TX_OUT idles high, drops to 0 for exactly one clock (START), then presents the 8 data bits of 0xA5 LSB-first (1,0,1,0,0,1,0,1), then the even parity bit, then returns high for STOP, then idles high again while Busy returns to 0.

With Do files : I learned some tricks from AI to make it more readable s



5. Results

All 5 test scenarios passed with 0 errors (see log above), confirming the design meets every requirement in the specification: correct frame format for all 3 parity configurations, correct Busy behavior, correct rejection of new data while busy, and correct asynchronous reset behavior.

6. Conclusion

The UART_TX block was implemented in Verilog following the recommended 4-block partition (Serializer, Parity Calc, FSM, MUX), verified with a self-checking testbench at 200 MHz, and all functional requirements from the specification were met.